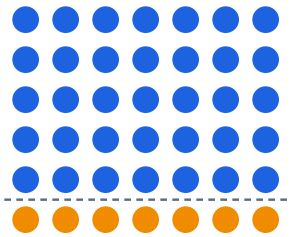
# Multiplication Strategies

Break a hard fact into facts you know.

### 6 × 7: 5 rows and 1 more row



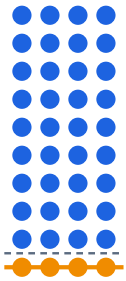
$5 \times 7 = 35$

$1 \times 7 = 7$

$35 + 7 = 42$

×5 is easy: half of ×10.
Then add one more row.

### 9 × 4: 10 rows, then take one away



$10 \times 4 = 40$

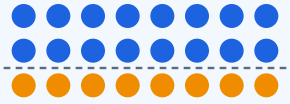
$40 - 4 = 36$

The crossed-out row
is the group taken away.

### A strategy for every fact

- **×3** = ×2 and one more group
- **×4** = double, then double again
- **×6** = ×5 and one more group
- **×7** = ×5 and ×2
- **×8** = ×4, then double
- **×9** = ×10, take one group away

### Your turn!



$2 \times 8 =$ 16

$1 \times 8 =$ 8

16 + 8 = 24

**For teachers**

Goal: use ×2, ×5 and ×10 to find the other facts by breaking an array apart (3.OA.5, 3.OA.7). Draw the dashed line, solve each part, then add (or take away for ×9). Practice a few facts a day until they are quick.

# Multiplication Strategies: Practice

Break the array. Solve each part. Put them together.

**Part 1** Times 3 ×3 = ×2 and one more group.

1) $3 \times 6 = 18$



$2 \times 6 = 12$

$1 \times 6 = 6$

$12 + 6 = 18$

2) $3 \times 4 = 12$



$2 \times 4 = 8$

$1 \times 4 = 4$

$8 + 4 = 12$

3) $3 \times 9 = 27$



$2 \times 9 = 18$

$1 \times 9 = 9$

$18 + 9 = 27$

4) $3 \times 7 = 21$



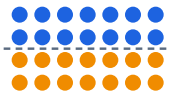
$2 \times 7 = 14$

$1 \times 7 = 7$

$14 + 7 = 21$

**Part 2** Times 4 ×4 = double, then double again.

5) $4 \times 7 = 28$



$2 \times 7 = 14$

$2 \times 7 = 14$

$14 + 14 = 28$

6) $4 \times 3 = 12$



$2 \times 3 = 6$

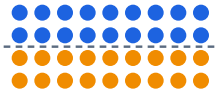
$2 \times 3 = 6$

$6 + 6 = 12$

## Multiplication Strategies: Practice (continued)

### Part 2 Times 4 (continued)

7) $4 \times 9 = 36$



$2 \times 9 = 18$

$2 \times 9 = 18$

$18 + 18 = 36$

8) $4 \times 6 = 24$



$2 \times 6 = 12$

$2 \times 6 = 12$

$12 + 12 = 24$

### Part 3 Times 6 ×6 = ×5 and one more group.

9) $6 \times 4 = 24$

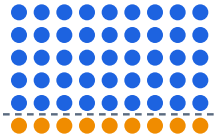


$5 \times 4 = 20$

$1 \times 4 = 4$

$20 + 4 = 24$

10) $6 \times 9 = 54$

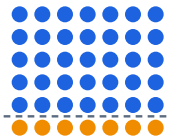


$5 \times 9 = 45$

$1 \times 9 = 9$

$45 + 9 = 54$

11) $6 \times 7 = 42$



$5 \times 7 = 35$

$1 \times 7 = 7$

$35 + 7 = 42$

12) $6 \times 3 = 18$



$5 \times 3 = 15$

$1 \times 3 = 3$

$15 + 3 = 18$

## Multiplication Strategies: Practice (continued)

**Part 4** Times 9 ×9 = ×10, take one group away.

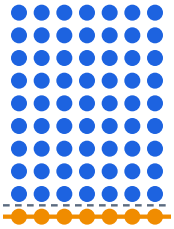
13) $9 \times 3 = 27$



$10 \times 3 = 30$

$30 - 3 = 27$

14) $9 \times 7 = 63$



$10 \times 7 = 70$

$70 - 7 = 63$

15) $9 \times 4 = 36$



$10 \times 4 = 40$

$40 - 4 = 36$

**Part 5** Times 8 ×8 = ×4, then double.

16) $8 \times 3 = 24$

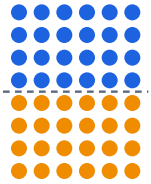


$4 \times 3 = 12$

$4 \times 3 = 12$

$12 + 12 = 24$

17) $8 \times 6 = 48$



$4 \times 6 = 24$

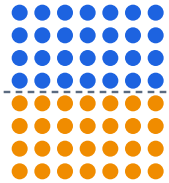
$4 \times 6 = 24$

$24 + 24 = 48$

## Multiplication Strategies: Practice (continued)

### Part 5 Times 8 (continued)

18) $8 \times 7 = 56$



$4 \times 7 = 28$

$4 \times 7 = 28$

$28 + 28 = 56$

### Part 6 Times 7 ×7 = ×5 and ×2.

19) $7 \times 4 = 28$

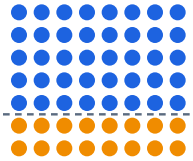


$5 \times 4 = 20$

$2 \times 4 = 8$

$20 + 8 = 28$

20) $7 \times 8 = 56$



$5 \times 8 = 40$

$2 \times 8 = 16$

$40 + 16 = 56$

21) $7 \times 7 = 49$



$5 \times 7 = 35$

$2 \times 7 = 14$

$35 + 14 = 49$

### Part 7 Times 3 No help this time.

22) $3 \times 7 = 21$

23) $7 \times 3 = 21$

24) $3 \times 9 = 27$

25) $9 \times 3 = 27$

26) $3 \times 6 = 18$

27) $4 \times 3 = 12$

## Multiplication Strategies: Practice (continued)

### Part 7 Times 3 (continued)

28) $3 \times 8 = 24$

29) $8 \times 3 = 24$

30) $3 \times 4 = 12$

31) $6 \times 3 = 18$

32) $3 \times 3 = 9$

### Part 8 Times 4 No help this time.

33) $4 \times 7 = 28$

34) $6 \times 4 = 24$

35) $4 \times 9 = 36$

36) $7 \times 4 = 28$

37) $8 \times 4 = 32$

38) $4 \times 8 = 32$

39) $9 \times 4 = 36$

40) $4 \times 4 = 16$

41) $4 \times 6 = 24$

### Part 9 Times 6 No help this time.

42) $9 \times 6 = 54$

43) $6 \times 8 = 48$

44) $7 \times 6 = 42$

45) $6 \times 6 = 36$

46) $8 \times 6 = 48$

47) $6 \times 7 = 42$

48) $6 \times 9 = 54$

### Part 10 Times 7, 8 and 9 No help this time.

49) $8 \times 7 = 56$

50) $7 \times 7 = 49$

51) $9 \times 8 = 72$

52) $8 \times 9 = 72$

53) $9 \times 9 = 81$

54) $7 \times 8 = 56$

## Multiplication Strategies: Practice (continued)

### Part 10 Times 7, 8 and 9 (continued)

55) $9 \times 7 =$ 63

56) $7 \times 9 =$ 63

57) $8 \times 8 =$ 64

### Part 11 Remember? Problems from earlier packets.

58) 6 groups of 5 = 30

59) half past 2 = 2:30

60) 3 dimes + 4 nickels = 50¢

61) 8 groups of 2 = 16

62) $502 - 267 =$ 235

63) 45 cm + 38 cm = 83 cm

### Part 12 Word problems Find the groups and how many in each.

64) A garden has 6 rows with 7 carrots in each row. How many carrots are there?

Number sentence 6 × 7 = 42

Answer 42 carrots

Draw or show your work

65) A spider has 8 legs. How many legs do 4 spiders have?

Number sentence 4 × 8 = 32

Answer 32 legs

Draw or show your work

66) A pack has 9 markers. How many markers are in 3 packs?

Number sentence 3 × 9 = 27

Answer 27 markers

Draw or show your work

## Multiplication Strategies: Practice (continued)

### Part 12 Word problems (continued)

67) There are 7 days in a week. How many days are in 8 weeks?

Number sentence $8 \times 7 = 56$

Answer 56 days

Draw or show your work

68) Challenge: A theater has 9 rows of 8 seats. 15 seats are empty. How many seats are full?

Number sentence $9 \times 8 = 72,\ 72 - 15 = 57$

Answer 57 seats

Draw or show your work

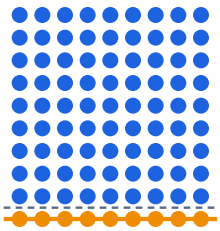
# Check-In: Multiplication Strategies

Show what you know. No help this time.

Mastery: 9 of 11 (80%)

## Part 1 Break it apart

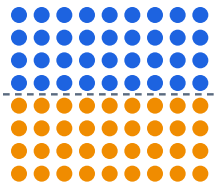
1) $9 \times 9 = 81$



$10 \times 9 = 90$

$90 - 9 = 81$

2) $8 \times 9 = 72$



$4 \times 9 = 36$

$4 \times 9 = 36$

$36 + 36 = 72$

## Part 2 Multiply

3) $3 \times 6 = 18$

4) $9 \times 9 = 81$

5) $7 \times 3 = 21$

6) $4 \times 9 = 36$

7) $7 \times 6 = 42$

8) $6 \times 6 = 36$

9) $3 \times 3 = 9$

10) $6 \times 9 = 54$

11) $7 \times 7 = 49$